Linear Algebra - Written exam - 10 February 2026 - Duration: 120 minutes - Total 30 points

Exercise 1 (5 pts). State the rank-nullity theorem and apply it to f: R^4 -> R^3, f(x)=Mx with M given.

Exercise 2 (9 pts). Diagonalize C = [[1,2,0],[2,1,0],[0,0,3]] with an orthogonal matrix.

Exercise 3 (8 pts). Find the kernel and image of the linear map $g(x,y,z)=(x+y, y+z, x-z)$.

Exercise 4 (8 pts). Prove that eigenvectors associated with distinct eigenvalues are linearly independent.